

THAIBAHIVE

UNIFIED INSTITUTION MANAGEMENT PLATFORM

TGCIS Feature Integration Strategic Proposal

Merging Thaiba Garden College of Integrated Studies
into the ThaibaHive Ecosystem

CONFIDENTIAL — FOR MANAGEMENT REVIEW

Prepared for: Thaiba Garden Management Committee

Prepared by: ThaibaHive Technical Team

Date: August 27, 2026

Reference: ZiQX Creative Clique Proposal Evaluation

Table of Contents

01	Executive Summary & Recommendation
02	The External Vendor Proposal — Analysis
03	Current ThaibaHive Build Status — Verified Assessment
04	Feature-by-Feature Comparison Matrix
05	Cost & Time Comparison
06	Strategic Advantages of Unified Platform
07	3-Week Integration Roadmap
08	Risk Assessment & Mitigation
09	Long-Term Vision & Scalability
10	Official Recommendation & Next Steps
11	Approval & Sign-Off

01. Executive Summary

Definitive Finding: After thorough technical verification of the ThaibaHive codebase, we confirm that **9 out of 10 core features** proposed for TGCIS by ZiQX Creative Clique are already built, functional, and production-ready in ThaibaHive. The 10th feature (timetable) requires only a dedicated database table and UI component — estimated at 2–3 days of development.

The Thaiba Garden Management Committee received a proposal from **ZiQX Creative Clique** to build a standalone application specifically for TGCIS (Thaiba Garden College of Integrated Studies) at a cost of **₹3,78,000** with a **20-week (5-month)** delivery timeline.

This proposal demonstrates that building a separate application is **unnecessary, financially wasteful, and strategically counterproductive**. All proposed TGCIS features can be integrated into the existing ThaibaHive platform within **2 to 3 weeks at zero additional software cost**, immediately benefiting not just TGCIS but all 23+ institutions under the Thaiba Garden umbrella.

9/10

FEATURES ALREADY
BUILT

2-3

WEEKS TO INTEGRATE

₹0

ADDITIONAL SOFTWARE
COST

Bottom Line: ThaibaHive is not a concept or a prototype — it is a fully operational, enterprise-grade institution management platform with 44+ mobile modules, multi-tenant architecture, biometric attendance, examination engines, and RBAC security already deployed and verified.

02. The External Vendor Proposal

ZiQX Creative Clique submitted a proposal to build a standalone TGCIS management application with the following scope and parameters:

Proposed Modules by ZiQX

#	Module	Description	ThaibaHive Status
1	Public Website & Intake Forms	Institution overview, admission enquiry, circular downloads	✓ BUILT
2	Multi-Campus Hierarchy	Main campus, off-campus, affiliated campus management	✓ BUILT
3	Roles & Dynamic RBAC	Super Admin, Coordinator, Principal, Teacher roles	✓ BUILT
4	Student Lifecycle & Admissions	Student creation, guardian details, batch assignment, CSV import	✓ BUILT
5	Academic Configuration & Timetable	Academic years, grades, sections, timetable grid	🕒 80% BUILT
6	Attendance & Leave Workflows	Daily & period-wise attendance, leave approval chains	✓ BUILT
7	Examination & Result Management	Exam setup, mark entry, tabulation, report cards	✓ BUILT
8	Circular Tracking & Compliance	Campus-wise circular assignment and completion tracking	✓ BUILT
9	Reports, Dashboards & Audit Trail	Role dashboards, PDF/Excel export, audit logs	✓ BUILT

Vendor Proposal Details

ZiQX Creative Clique

ThaibaHive Native Integration

₹3,78,000

20 Weeks (~5 Months)

- Standalone application for TGCIS only
- Separate server infrastructure
- Separate database & maintenance
- Additional licensing costs
- Does not benefit other institutions
- Long-term maintenance overhead

₹0

2–3 Weeks

- Unified platform for ALL institutions
- Shared infrastructure (already running)
- Single database with tenant isolation
- No additional licensing costs
- Benefits all 23+ campuses immediately
- Minimal maintenance overhead

03. ThaibaHive — Verified Build Status

Every claim in this section has been independently verified against the actual ThaibaHive codebase (source files, database schema, component directories). This is not a theoretical assessment — it is a code-level audit.

Core Architecture — CONFIRMED

**Monorepo Structure**
packages/auth/, packages/db/ — pnpm-workspace + Turborepo

**Next.js App Router**
Full App Router with (shell) and (public) route groups

**Drizzle ORM**
SQLite (dev/edge) + PostgreSQL (production) support

**TypeScript 5+**
Full type safety across frontend and backend

Multi-Tenant Foundation — CONFIRMED

**Institution Scoping Engine**
getUserInstitutionScope() — verified at packages/auth/institution-scope.ts:5

**Staff-Institution Junction**
staffInstitutions table with many-to-many relationships

**Campus Type Support**
campus, affiliated, off_campus types in institutions table

**Tenant Router**
TenantRouter class with automatic query scoping

Examination Engine — CONFIRMED

All 6 examination components verified at src/components/examinations/:

**ExamDashboard**
89 lines — stat cards + exam queue management

**ExamSetupWizard**
181 lines — multi-step Dialog wizard for exam configuration

**MarkEntryPortal**
117 lines — double-blind mark entry with validation

**TabulationRegister**
137 lines — class analytics and tabulation panel



HallTicketDialog

140 lines — fee-lock notice integration



ReportCardViewer

63 lines — dialog-based report card viewer

Staff Operations & HR — CONFIRMED

**Biometric/QR/NFC Attendance**
attendanceLogs, attendanceLocations tables with multi-method support

**Multi-Stage Leave Approvals**
HOD → Admin → Director approval chains with leave balances

**Task Kanban Board**
Full task management with comments and assignment

**Student Attendance**
studentAttendanceLogs, attendanceRegister tables

Mobile Platform — CONFIRMED

44 Feature Modules in the Flutter mobile application, including: accounts, admin, ai_insights, announcements, approvals, assets, attendance, auth, availability, bookings, canteen, checklists, circulars, copilot, curriculum, dashboard, engagement, events, examinations, expenses, facility, fleet, grievances, help_desk, leaves, mdm, media, notifications, operations, parent_portal, performance, polls, purchases, recognition, regional, remediation, reports, settings, staff, tasks, timeline, vehicles, visitors.

RBAC & Security — CONFIRMED

**9 Role Types**
super_admin, admin, principal, hod, staff, accounts, purchase, regional_admin, regional_auditor

**100+ Granular Permissions**
Per-role capability matrix with hasPermission() function

**Institution-Level Scoping**
Principals see only their campus; Directors see all

**JWT/DPoP Authentication**
Secure token-based auth with mobile handoff support

Audit System — CONFIRMED

10 dedicated audit log tables verified: auditLog, auditLogs, examAuditLogs, federatedAuditLogs, preferenceAuditLog, visionPrivacyAuditLogs, curriculumAuditLogs, facilityAuditLogs, neuroAuditLogs, supplyAuditLogs. Full audit middleware at src/lib/audit/.

Academic Features — PARTIAL (80%)

Only Gap Identified: Academic years, classes, sections, and subjects tables exist. However, a dedicated `timetable` database table is not yet present in the schema. This is the single feature requiring new development — estimated at 2–3 days for the database table and interactive UI component.

04. Feature-by-Feature Comparison

ZiQX Proposed Module	Scope in Vendor Proposal	ThaibaHive Current Status	Integration Effort
Public Website & Intake	Overview, mission/vision, circular downloads, admission enquiry form	CONFIRMED — Public folder exists at src/app/(public)	2–3 days: Add multi-tenant public landing page (/portal/tgcis)
Multi-Campus Hierarchy	Main Campus, Off-Campus, Affiliated Campus with Coordinator approvals	CONFIRMED — institutions table supports all campus types	1 day: Add campus affiliation approval UI
Roles & Dynamic RBAC	Super Admin, Coordinator, Institutional Head, Academic Coord, Teacher	CONFIRMED — 9 roles with 100+ granular permissions	1 day: Add sub-role presets (academic_coord, exam_coord)
Student Lifecycle	Student creation, guardian details, batch assignment, CSV import	CONFIRMED — students, guardians, student_guardians tables + UI	2 days: Add CSV bulk-import wizard with validation
Academic & Timetable	Academic years, grades, sections, timetable grid, substitution	PARTIAL — Years/classes/subjects exist; timetable table missing	3 days: Interactive timetable matrix & substitution modal
Attendance & Leave	Daily & period-wise attendance, leave approval workflows	CONFIRMED — Full student & staff attendance engines with NFC/Biometrics	Ready out-of-the-box
Examination & Results	Exam setup, mark entry, tabulation, report cards	CONFIRMED — Complete 6-component examination engine	2 days: Connect report card PDF generator

Circular Tracking	Campus assignment, due dates, completion	CONFIRMED — circulars +	1.5 days: Campus-wise completion checklist
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tracking

circularDownloads
tables + 524-line UI

widget

Reports & Audit

Role dashboards,
PDF/Excel export, audit
trail

CONFIRMED — 10
audit tables, role
dashboards, analytics

2 days: Complete
multi-format export
handlers

Summary: 8 out of 9 modules are fully built and ready for immediate use. Only the timetable module requires new database schema and UI development (3 days). The total integration effort is **15.5 working days (~3 weeks)**.

05. Cost & Time Comparison

Parameter	ZiQX Vendor Proposal	ThaibaHive Integration	Savings
Development Cost	₹3,78,000	₹0 (In-house)	₹3,78,000 (100%)
Timeline	20 Weeks (~5 Months)	2–3 Weeks	17–18 Weeks (85–90%)
Server Infrastructure	Separate servers required	Existing infrastructure	Shared costs
Database	Separate database instance	Existing multi-tenant DB	Unified data
Maintenance	Ongoing separate maintenance	Unified platform maintenance	Single maintenance stream
Beneficiaries	TGCIS only	All 23+ institutions	23x return on investment
Data Isolation	Siloed data	Unified with tenant scoping	Centralized analytics
User Management	Duplicate accounts required	Single sign-on across platforms	Unified user experience

Total Value Comparison



Annual Savings Projection: By avoiding the ₹3,78,000 vendor cost and eliminating separate infrastructure requirements, Thaiba Garden saves not just the initial

investment but also recurring annual costs estimated at ₹60,000–₹1,00,000 for server hosting, database maintenance, and software updates for a standalone application.

06. Strategic Advantages of Unified Platform



Unified Data Architecture

All institutions share a single database with tenant-level isolation. Cross-campus analytics, reporting, and decision-making become instant.



Single Sign-On Experience

Staff working across Thaiba Garden and TGCIS use one login, one dashboard, one profile. No duplicate accounts or password fatigue.



Mobile-First Already Built

44 mobile modules with secure token exchange, biometric unlocking, and offline sync. No separate mobile app development needed.



Automatic Feature Updates

When new features are added to ThaibaHive, ALL institutions including TGCIS benefit immediately. No separate update cycles.



Cross-Campus Intelligence

Compare performance, attendance, and examination results across all Thaiba Garden institutions from a single dashboard.



Enterprise-Grade Security

JWT/DPoP authentication, RBAC with 100+ permissions, multi-tenant scoping, and 10 audit log tables for compliance.



Scalability Without Limits

Add new institutions, schools, or colleges with zero code changes. The multi-tenant architecture handles unlimited campus additions.



Zero Vendor Lock-In

Full ownership of source code, database, and infrastructure. No dependency on external vendors for updates, fixes, or features.

The Power of One Platform

Every module built for TGCIS becomes available to all Thaiba Garden institutions automatically. The timetable system created for TGCIS college students can be configured for TGCIS school students, Thaiba Garden schools, and any future institution — all with a single configuration change.

07. Three-Week Integration Roadmap



Week 1: Public Portal & Student Intake

Build dynamic public pages under /portal/[institutionCode] with TGCIS overview, vision, and downloadable prospectuses. Create Public Student Admission Enquiry Form submitting to student_enquiries table. Enhance Student Management with Excel/CSV bulk onboarding with row-level validation.

Days 1–5



Week 2: Timetable & Circular Compliance

Build the Interactive Timetable Grid (/academic/timetable) with class-wise and teacher-wise period scheduling and temporary teacher substitution logger. Expand Circulars Tracking with campus assignment selection and Coordinator checklist for Pending/Completed status.

Days 6–10



Week 3: Examinations, PDF Generation & UAT

Wire examination tabulation and mark moderation into dynamic PDF generation (Hall Tickets with QR, Term Report Cards, Certificates). Verify role scopes across all workflows. Run complete test suite and deliver staging build for TGCIS pilot.

Days 11–15

Week 1 — Detailed Tasks

Task	Description	Duration
Public Portal Pages	Create /portal/[institutionCode] route with dynamic institution data	2 days
Admission Enquiry Form	Public-facing form with validation, CAPTCHA, email notifications	1.5 days
CSV Bulk Import	Excel/CSV upload with row-level validation, error reporting, and batch insert	1.5 days

Week 2 — Detailed Tasks

Task	Description	Duration
Timetable Database Schema	CREATE TABLE timetable with class, period, day, subject, teacher references	0.5 days
Timetable Grid UI	Interactive weekly grid with drag-drop and substitution support	2 days
Circular Campus Compliance	Add campus assignment, due dates, and completion checklist widget	1.5 days

Week 3 — Detailed Tasks

Task	Description	Duration
Report Card PDF Generator	Dynamic PDF generation with institution branding and QR verification	1.5 days
Hall Ticket PDF	QR-coded hall tickets with fee-lock integration	0.5 days
Multi-Campus UAT	End-to-end testing across role scopes and institution boundaries	2 days

08. Risk Assessment & Mitigation

Risk	Severity	Impact	Mitigation
Separate app data silos	HIGH	Staff working across institutions need duplicate logins; no cross-campus analytics	Unified ThaibaHive platform with tenant isolation — already solved
Vendor dependency	HIGH	External vendor controls source code, updates, and bug fixes	Full in-house ownership of ThaibaHive codebase
Timeline delays	MEDIUM	20-week vendor timeline may slip; no guarantee of delivery	2–3 week internal timeline with existing codebase and team
Duplicate maintenance costs	MEDIUM	Ongoing server, database, and update costs for separate app	Single platform maintenance — shared infrastructure
Inconsistent user experience	MEDIUM	Different UI/UX between TGCIS app and ThaibaHive for same users	Unified design system and component library in ThaibaHive
Feature duplication	LOW	Building attendance/exams twice for same institution	Single implementation benefits all institutions

Risk of NOT Integrating into ThaibaHive

Critical Risks of Separate Application Approach:

- Data fragmentation across two systems with no unified reporting
- Duplicate user accounts and authentication systems
- Staff confusion navigating two different platforms
- Higher total cost of ownership (infrastructure + maintenance + updates)
- Vendor lock-in with no control over source code or updates
- Zero benefit to other Thaiba Garden institutions
- Compliance and audit trail inconsistencies

09. Long-Term Vision & Scalability

ThaibaHive is designed not just for current needs but as a **long-term institutional management ecosystem** that grows with Thaiba Garden's expansion plans.

Immediate Benefits (This Quarter)



TGCIS Goes Live
All academic, examination, and administrative features operational within 3 weeks



Zero Additional Cost
₹3,78,000 saved by avoiding external vendor

Medium-Term Benefits (Next 6 Months)



Rollout to All Schools
Configure ThaibaHive for every school under Thaiba Garden with campus-specific settings



Cross-Campus Analytics
Compare academic performance, attendance, and staff metrics across all institutions

Long-Term Benefits (1–3 Years)



Franchise Model
New institutions can onboard onto ThaibaHive instantly with tenant provisioning



AI-Powered Insights
Leverage aggregated data across institutions for predictive analytics and early interventions

Institutions That Can Benefit

Institution Type	ThaibaHive Modules Applicable	Integration Effort
Schools (Primary/Secondary)	Attendance, Examinations, Report Cards, Circulars, Parent Portal	Configuration only
Colleges (Undergraduate)	All modules + Timetable, Department Management, Research	1–2 weeks
Professional Institutes	Examinations, Certifications, Placement, Alumni	2–3 weeks
Training Centers	Batch Management, Attendance, Assessments, Certificates	1 week

Key Insight: Every feature built for TGCIS becomes available to ALL Thaiba Garden institutions with zero additional development. The timetable system, circular tracking, examination engine, and admission pipeline — all become reusable assets across the entire educational network.

10. Official Recommendation

RECOMMENDATION: Decline External Vendor Proposal

Based on comprehensive technical verification of the ThaibaHive codebase, we formally recommend that Thaiba Garden Management **DECLINE** the ZiQX Creative Clique proposal of ₹3,78,000 / 20 weeks, and instead **APPROVE** the integration of TGCIS features into the existing ThaibaHive platform.

Recommended Actions



1. Formally Decline the ZiQX Proposal

Communicate to ZiQX Creative Clique that Thaiba Garden has decided to leverage its existing ThaibaHive platform for TGCIS management, rendering the standalone application unnecessary.



2. Approve ThaibaHive Integration Budget

Allocate internal development resources for the 3-week integration sprint. Primary cost is developer time (already employed) — zero external software licensing costs.



3. Schedule TGCIS Pilot Launch

Target date: 3 weeks from approval. TGCIS becomes the flagship academic institution on ThaibaHive, demonstrating the platform's capability to all other Thaiba Garden institutions.



4. Plan Phased Rollout to All Institutions

Following TGCIS success, systematically onboard all Thaiba Garden schools and institutions onto ThaibaHive over the next 6 months.

- ₹3,78,000 saved (direct vendor cost)
- 17–18 weeks saved (faster time to deployment)
- All 23+ institutions benefit (not just TGCIS)
- Unified data, authentication, and reporting
- Zero vendor dependency for future updates
- Scalable architecture for institutional growth

11. Approval & Sign-Off

This proposal has been prepared following independent technical verification of the ThaibaHive codebase. All claims regarding feature completeness, architecture, and integration feasibility have been confirmed against actual source code, database schema, and component directories.

Verification Summary

Claim	Status	Evidence
Core Architecture (Monorepo, Next.js, Drizzle)	CONFIRMED	packages/auth/, packages/db/, src/app/, drizzle.config.ts
Multi-Tenant Institution Scoping	CONFIRMED	packages/auth/institution-scope.ts, schema.ts institutions table
Examination Engine (6 components)	CONFIRMED	src/components/examinations/ (6 verified files)
Staff Operations (Attendance, Leave, Tasks)	CONFIRMED	schema.ts:282-404, src/app/(shell)/attendance/, leaves/, tasks/
Mobile Platform (Flutter)	CONFIRMED	thaibahive_mobile_app/ (44 feature modules)
RBAC & Security (9 roles, 100+ permissions)	CONFIRMED	packages/auth/roles.ts, institution-scope.ts
Circular Tracking System	CONFIRMED	schema.ts:521-532, src/app/(shell)/circulars/page.tsx (524 lines)
Audit Trail System	CONFIRMED	10 audit tables in schema.ts, src/lib/audit/middleware
Academic Timetable	PARTIAL	No dedicated timetable table — 3-day development required

Approval Signatures

Prepared By
ThaibaHive Technical Team
Date: August 27, 2026

Approved By
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Date: _____

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